

KARTHIKEYA APPASANI

Hyderabad, India ◊ +91 9676374416
[karthikeyappasani17@gmail.com] ◊ GitHub ◊ LinkedIn

SUMMARY

B.Tech Information Technology student (CGPA: 8.68/10) with experience building full-stack and AI-powered applications using Java, Python, SQL, Firebase, and cloud technologies. Developed projects in civic issue management and crop disease detection while strengthening problem-solving skills through 100+ DSA challenges on LeetCode. Interested in Software Engineering and Artificial Intelligence roles.

EDUCATION

MLR Institute of Technology, Hyderabad, India
Bachelor of Technology in Information Technology

Aug 2023 – May 2027
CGPA: 8.68/10

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C, SQL
Web Technologies: HTML, CSS, Next.js, Tailwind CSS
Databases: MySQL, PostgreSQL, Firebase Firestore
AI/ML & Backend: PyTorch, OpenCV, FastAPI
Tools & Platforms: Git, GitHub, AWS, Firebase
Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS

PROJECTS

CivicTrack

Next.js, React, Tailwind CSS, Firebase, Firestore, Leaflet, OpenStreetMap

- Developed a full-stack civic issue reporting and monitoring platform enabling citizens to report and track infrastructure issues including potholes, garbage accumulation, water supply disruptions, and streetlight failures.
- Implemented authentication, complaint management, image uploads, GPS-based location tracking, and real-time status updates using Firebase Authentication, Firestore, and Storage.
- Designed role-based dashboards for citizens and authorities with analytics and map-based visualization to improve issue monitoring and resolution workflows.

Path – AI Career Mentor

Next.js, TypeScript, React, Tailwind CSS, Firebase, Firestore, Groq SDK, Google Generative AI

- Built an AI-powered career mentorship web app using Next.js App Router and TypeScript, with Firebase Authentication (Google/GitHub OAuth), a guided onboarding flow, and persistent per-user chat sessions.
- Implemented a streaming AI mentor chat engine with a triple-fallback provider architecture (Gemini Flash, Groq Llama 3.3 70B, Ollama) driven by a dynamically generated system prompt personalized to each user's academic background, career goals, and skill level.
- Designed a free-tier-friendly architecture with daily message limits, Markdown-rendered responses via react-markdown, and a responsive Tailwind CSS interface built as a PWA.

AgriAid

PyTorch, FastAPI, OpenCV, PostgreSQL, AWS S3

- Collaborated in a team to develop an AI-powered crop disease detection platform using transfer learning on a pre-trained Convolutional Neural Network (CNN).
- Trained and evaluated the model on a Kaggle plant disease dataset, achieving approximately 96% classification accuracy across multiple disease categories.
- Developed FastAPI-based prediction services and integrated OpenCV, PostgreSQL, and AWS S3 for image processing, data management, and scalable storage.

ACHIEVEMENTS & ACTIVITIES

- Solved 100+ Data Structures and Algorithms problems on LeetCode.
- Participated in a Hackathon conducted by CVR College of Engineering, Dundigal.
- Participated in Microsoft ISOC UA Industry Visit and technical learning program.

CERTIFICATIONS

- AWS Cloud Practitioner Essentials
- Google Cloud Jams Certification
- Google Cloud Skill Badges